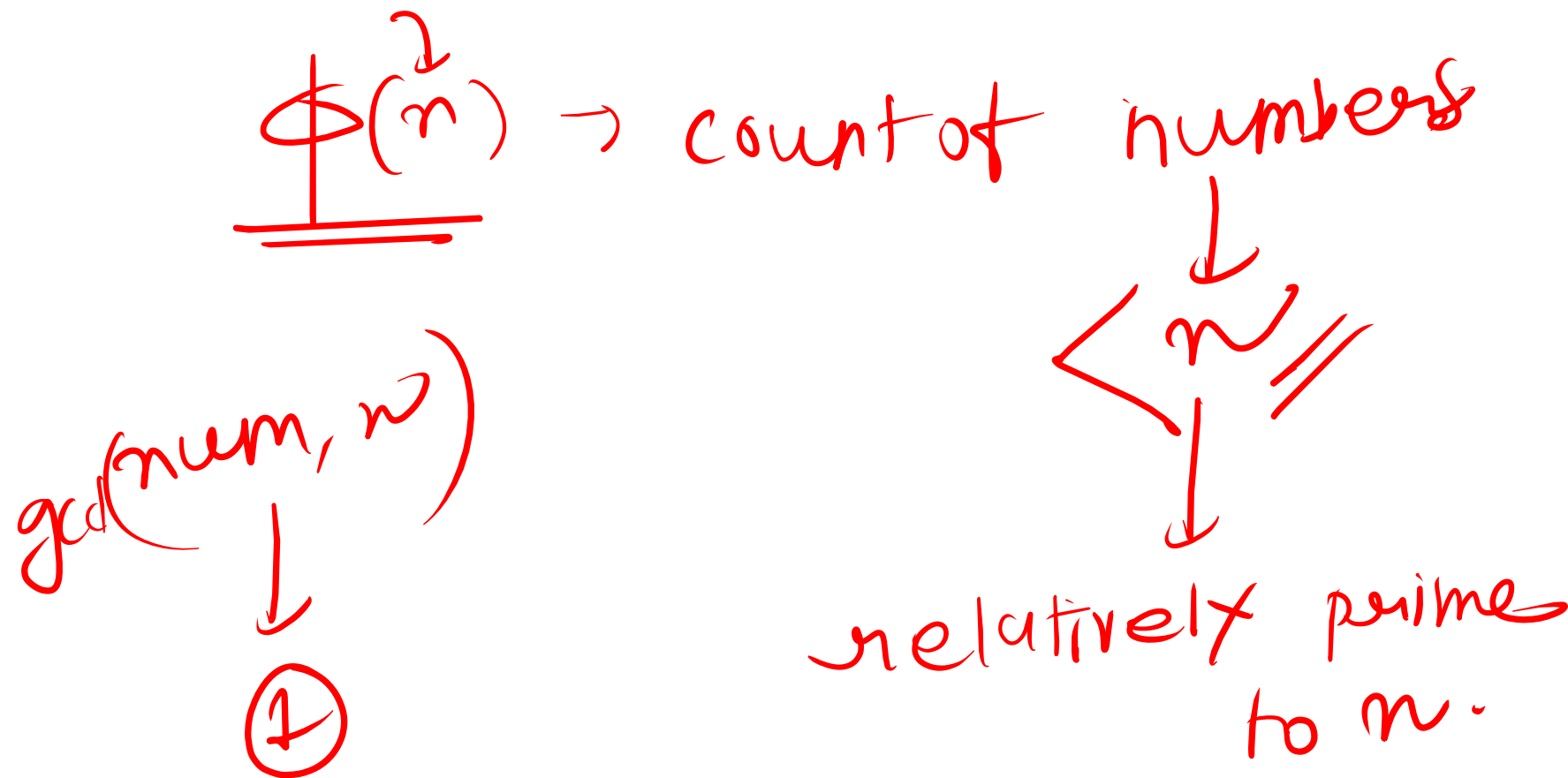


Euler Totient function.



$$n = 6$$

$$\underline{1\ 2\ 3\ 4\ 5} < 6$$

$$1\ 5\} \rightarrow \underline{\underline{2}}$$

$$\begin{array}{c} n = \text{prime} \\ \downarrow \\ \text{num} < n \\ \underbrace{\hspace{1cm}} \\ \text{gcd} = 1 \end{array}$$

$$\begin{array}{c} \cancel{1/13} \\ \nearrow \\ \textcircled{13} \rightarrow \\ \searrow \end{array}$$

$$\begin{array}{c} 1 \sim 12 \\ \underbrace{\hspace{1cm}} \\ \downarrow \\ 12 \checkmark \end{array}$$

$$N = \left\{ p_1^{e_1} p_2^{e_2} p_3^{e_3} \dots p_n^{e_n} \right\}$$

$$\prod_{i=1}^n p_i^{e_i}$$

$p_i \rightarrow i$ th prime factor

$e_i \rightarrow$ power of prime p within N .

$$\phi(n) =$$

$$N = \frac{1}{10} \left\{ \underline{2^1} * \underline{5^1} \right\}$$

$$\frac{18}{1} \left\{ 2^1 * 3^2 \right\}$$

$\{1, \dots, n\}$ num
 $\{\phi\}$ strategy?
 $\phi(n) = n * (1 - \frac{1}{p_2}) (1 - \frac{1}{p_2}) (1 - \frac{1}{p_3}) \dots (1 - \frac{1}{p_n})$

$$\phi(n) = n * \prod_{i=1}^n \left(\frac{p_i - 1}{p_i} \right)$$

$$n (1 - \frac{1}{p_i})$$

$$\phi(n) = n \prod_{i=1}^n \left(1 - \frac{1}{p_i} \right)$$

$i \rightarrow \text{prime}$
 $\phi(i) = \underline{i-1}$

$f[j] \rightarrow \text{reduced}$
 $i \rightarrow$

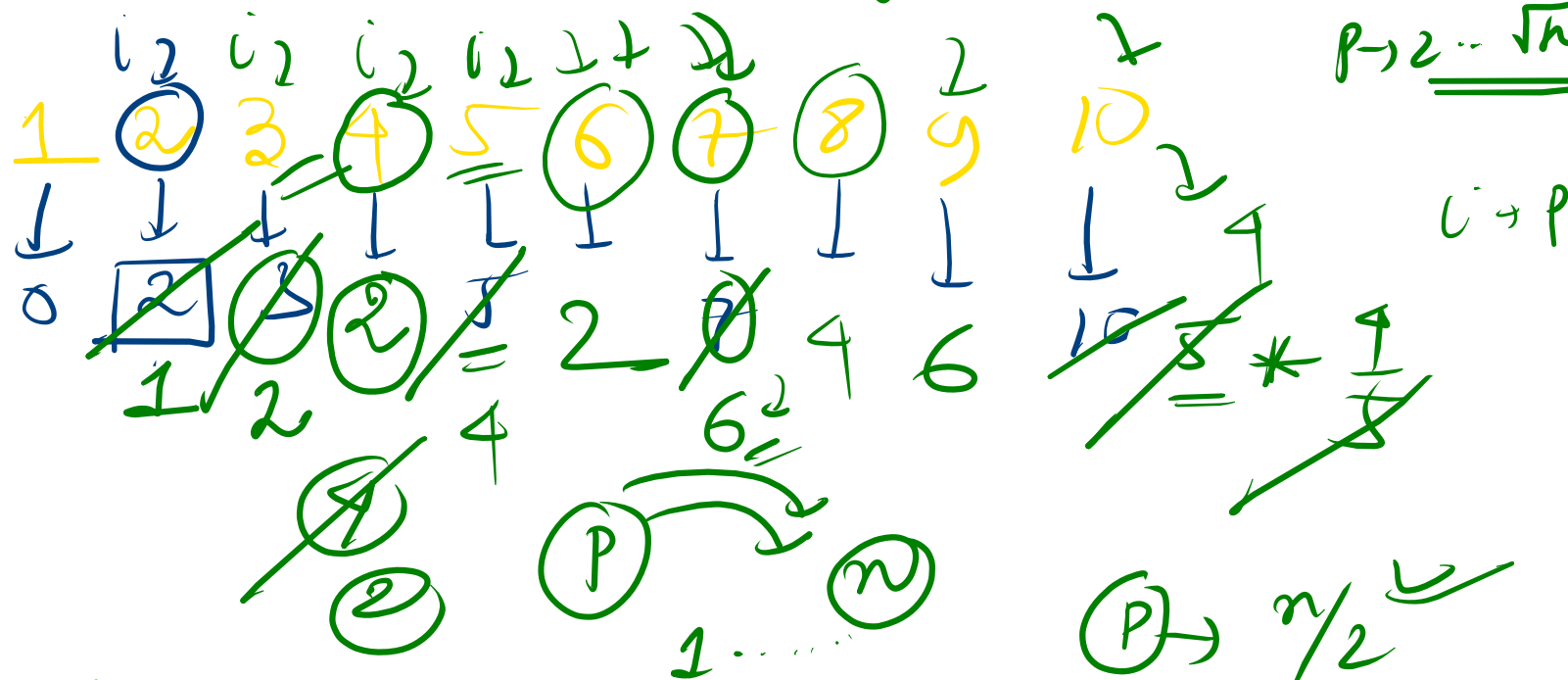
$$2 \rightarrow (1 - \frac{1}{2}) = \frac{1}{2}$$

$p \rightarrow \text{num}$

$$\text{num} * (1 - \frac{1}{p}) = \text{num} * (\frac{p-1}{p})$$

$$n \rightarrow n * (1 - \frac{1}{p})$$

Sieve ! $i \rightarrow 2 \dots n/2$ $p \rightarrow p^2$
 $n \geq 10$ $p \rightarrow 2 \dots \sqrt{n}$



$i \rightarrow \text{totient}[i] \rightarrow \checkmark$
2, (i) $\swarrow$
 $\rightarrow$

$$\text{totient}[n] \rightarrow \left\{ \frac{i}{i} \right\}$$

$$\text{totient}[1] = 0;$$

$\text{totient} \checkmark$

for ($i=2$; $i \leq n$; $i++$) {

if ($\text{totient}[i] == i$) {

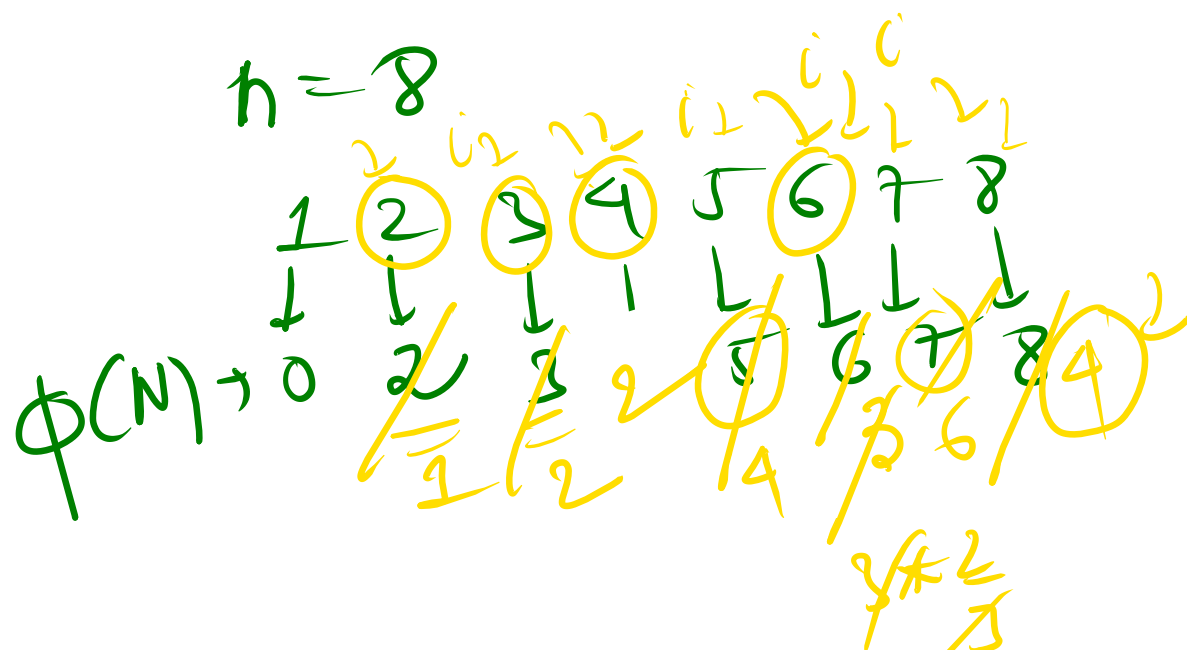
for ($j = i * 2$; $j \leq n$; $j += i$) {

$\text{totient}[j] *= (i-1);$

} $\text{totient}[j] /= i;$

$\text{totient}[i] = i-1; \}$

$\text{totient}[i] = i$?



$\textcircled{p}, \textcircled{2p}, 3p, 4p, \dots$

$$i \rightarrow \phi(i) = \textcircled{i}$$

$p \rightarrow \text{num}$

$$\phi \rightarrow \text{num} * \left(1 - \frac{1}{p}\right)$$



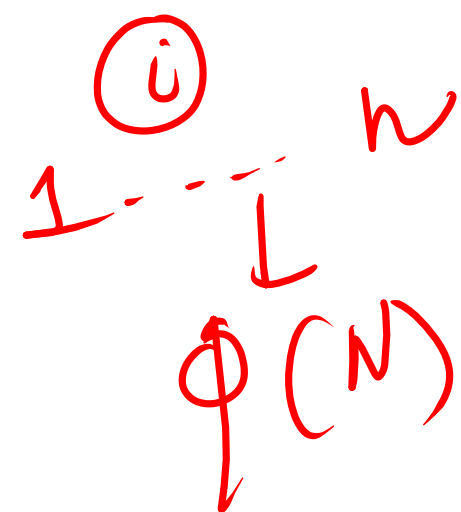
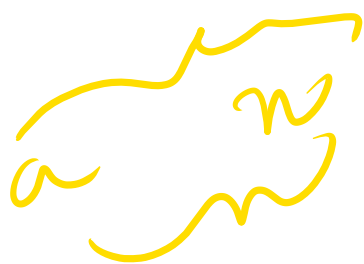
$\frac{1}{2}$

Euler's φ

$$a^{\phi(n)} \equiv 1 \pmod{n}$$



$a, n = \text{coprime}$.



$\text{num} \rightarrow \phi(\text{num})$

~~$\phi(\text{num})$~~

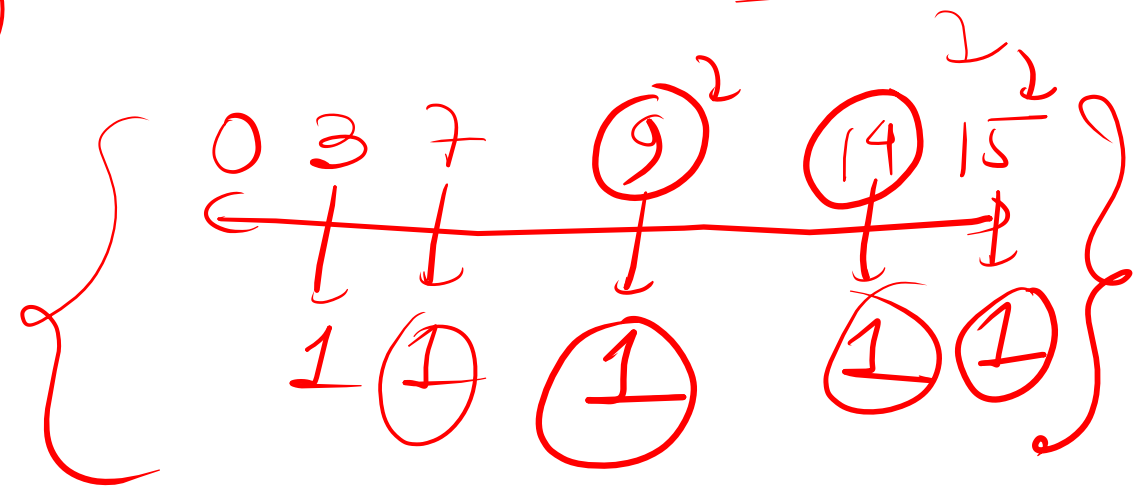
$\text{num} \left(1 - \frac{1}{p_i}\right) \dots$

$$\underline{\underline{3}} \quad \underline{\underline{14}} \quad \overset{2}{\textcircled{15}} \quad 7 \quad \underline{\underline{9^2}}$$

$$\overset{2}{(15, 15)} \rightarrow \textcircled{5}$$

$$\overset{2}{\dots} \overset{15}{\dots} \rightarrow \textcircled{14}$$

$$(14, 14)$$



$$\underline{\underline{16}} \quad \overset{1}{\text{freq}} \quad \underline{\underline{9}} \rightarrow \text{gcd}$$

$$\{1 \dots 15\} \rightarrow \text{gcd}$$

$$\textcircled{9}$$

$$g \rightarrow 15 \rightarrow 1$$

$$g \rightarrow 14 \rightarrow 1$$

$$\underline{\underline{g \rightarrow 9 \rightarrow 1}}$$

$$g \rightarrow 8 \rightarrow 0$$

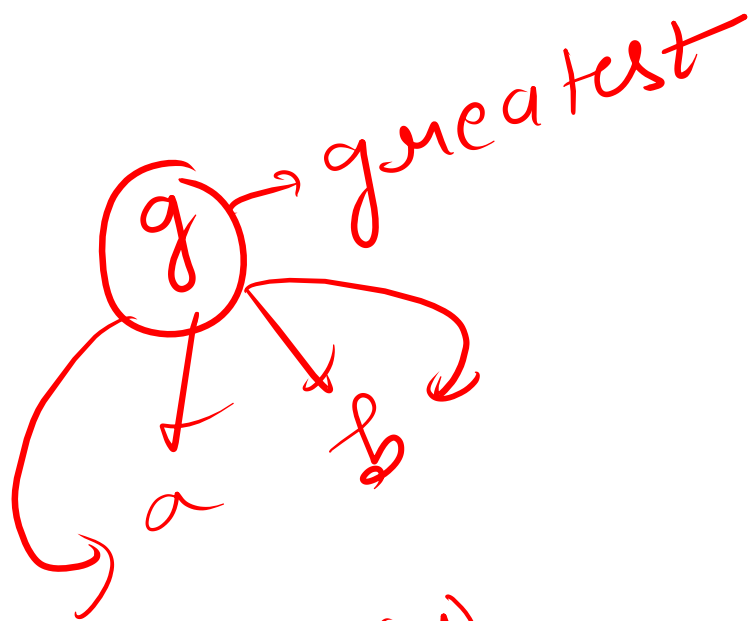
$$g \rightarrow \underline{\underline{7}} \rightarrow \textcircled{2}$$

$$\text{ans} \rightarrow \textcircled{7} \checkmark$$

$$(7, 14) \rightarrow \textcircled{7} \checkmark$$

$$(P_1, P_2) \rightarrow \textcircled{9}$$

$$\{1 \dots 15\}$$



(1 ... max) $\rightarrow$ gcd

$\sim$ (g) $\rightarrow$ > 1 multiple

$\hookrightarrow$ gcd ✓

1 (g) $\rightarrow$ $\log(M)$
 (M) $\rightarrow$ $\log(M)$
 (M $\log(M)$)

(1) ... (M)

$\approx \log(M)$

$M \rightarrow \left(\frac{M}{1}, \frac{M-1}{1}, \frac{M-2}{1}, \dots \right)^{M/2} \frac{M}{4} \frac{M}{8}$

1 $\rightarrow$ default
ans

1 2 3 4 ... M

$$\left\{ \frac{M}{2} \quad \frac{M}{3} \quad \frac{M}{4} \quad \dots \quad \frac{M}{M} = 1 \right\}$$

$(M) * \log(M)$

$\rightarrow \log(M)$

$$\underline{M} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{\textcircled{M}} \right) \rightarrow \log(M)$$

$$\{ M * \log(M) \} \quad M \rightarrow \text{max value.}$$

ans += $\sum_{i=1}^n \sum_{j=1}^i j$ multiples ✓ $n=5$ $O(n \log(n))$ $n \leq 10^6$

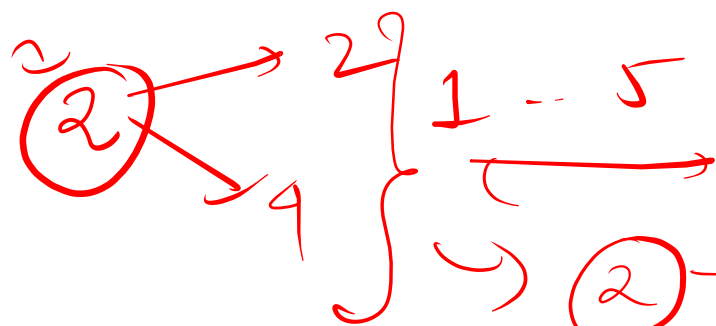
$\sum_{d=1}^n d * \text{term}$

$$\rightarrow (1) + (1+2) + (1+3) + (1+2+4) + (1+5) \rightarrow \underline{\underline{ans}}$$

$$\rightarrow \underline{1} * \underline{5} + \underline{2} * \underline{2} + \underline{3} * \underline{1} + \underline{4} * \underline{1} + \underline{5} * \underline{1}$$

✓ $\underline{5} + \underline{4} + \underline{3} + \underline{4} + \underline{5}$ $d \rightarrow 1 \dots n$

$$\text{sum} += \sum_{d=1}^n d * \left\lfloor \frac{n}{d} \right\rfloor$$



$\begin{matrix} 2 \\ \textcircled{4} \end{matrix} \rightarrow \text{no. factor}$

$1 \dots n$

$\frac{n}{2} = (2 * 2)$

$\underline{\underline{\text{ans}}} \leftarrow + \leftarrow \begin{matrix} 2 \\ \textcircled{4} \end{matrix}$

$2 \rightarrow 1 \dots n$

$\rightarrow \left(\frac{n}{2} \right) \rightarrow 2 \text{ factor}$

$2 + 2 + 2 + 2 \dots$

$\downarrow$
 $\textcircled{\frac{n}{2}} \rightarrow (+1)$

$+ 2 * \left[\frac{n}{2} \right] / 3 + \left[\frac{3 * n}{3} \right] \rightarrow \left[\frac{n}{4} \right] * 4$



10^{12}
 10^8

1 $\rightarrow 1 * n$
 2 $\rightarrow 2 * \lceil n/2 \rceil$

3 $\rightarrow 3 * \lceil n/3 \rceil$

$\vdots$
 n

$\rightarrow 1 \dots n$
 $n \rightarrow$

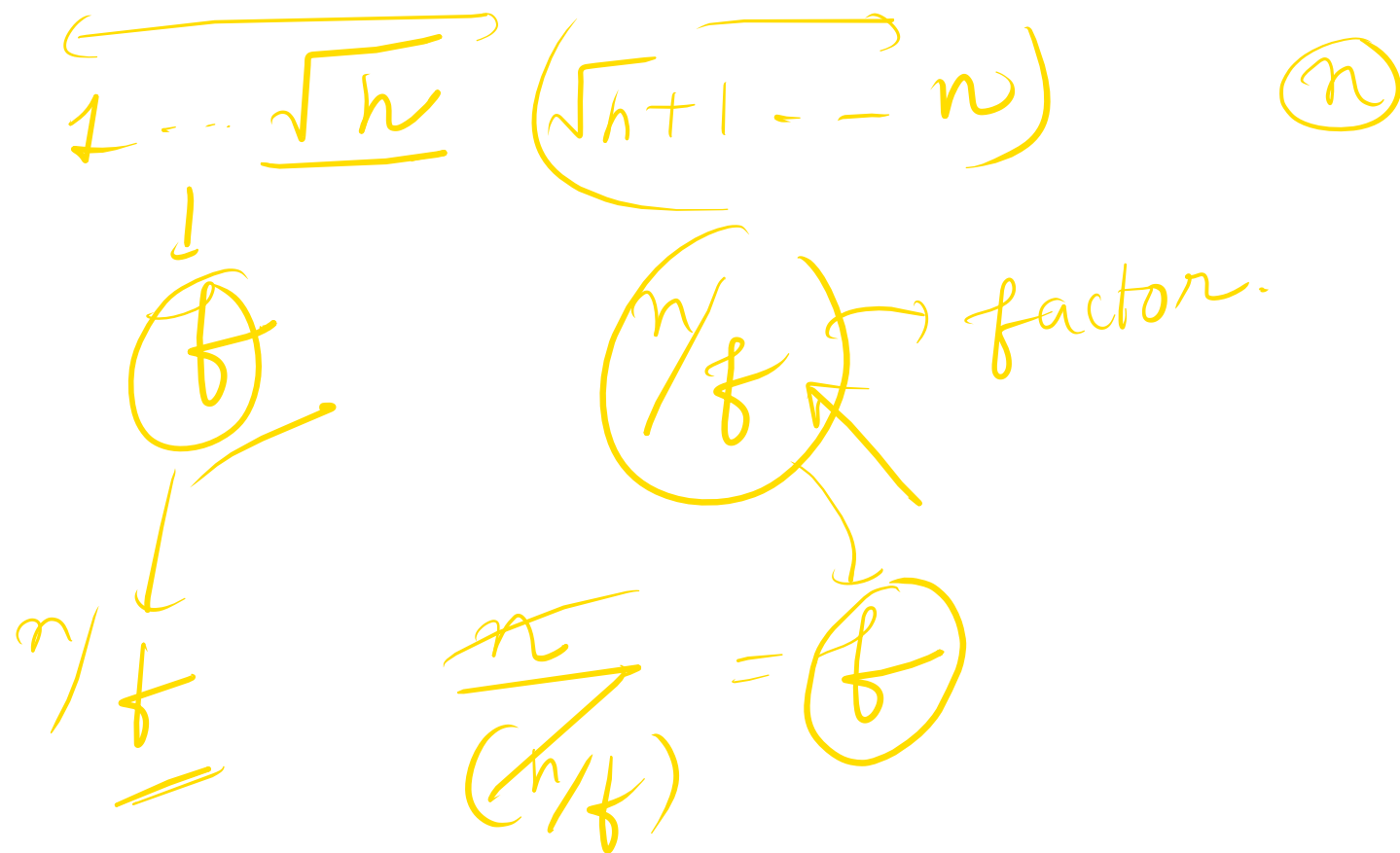
$(1 \dots n)$

$f \rightarrow + \rightarrow ans = f * \lceil n/b \rceil$
 $\hookrightarrow \text{int div.}$

$(1 \dots n) \rightarrow \underline{\text{factors}}$

$\sqrt{n} \rightarrow \boxed{10^6} \approx \leq 10^8$

$$(1 \dots \sqrt{n})$$



$$2 + \underline{f * [n/f]} = + \quad \leftarrow \text{ans} \quad + = \quad \underline{f * [n/f]} + \underline{[n/f] * f}$$

$$\underline{n=5} \rightarrow \underline{\sqrt{5}} = 2$$

$$\underline{1 \dots 5} \quad \text{⑤}^2$$

$$f \rightarrow 1 \dots \sqrt{n}$$

$$\text{ans} \rightarrow \left(\frac{f}{n/f} \right) \rightarrow 2 * f * \left(\frac{n}{f} \right)$$

$$1 \dots 2$$

$$\text{⑤}^2 \quad \begin{matrix} \downarrow \\ 2 \rightarrow \text{③}^2 \end{matrix}$$

$$1/5 \leftarrow f=1 \rightarrow 2 * 1 * \frac{5}{1} = 10$$

$$2/4 \leftarrow f=2 = 2 * 2 * \left[\frac{5}{2} \right] = 8$$

$$\leftarrow f=3 \rightarrow \frac{5}{3} = \underline{\underline{1 * 3}} = \underline{\underline{19}}$$

$$\begin{matrix} \downarrow \downarrow \\ 5/2 = \underline{\underline{\text{②}}} \end{matrix} \quad \text{③}^2$$

$$2^2 \text{③}^2$$

$$8 \frac{2}{5} \frac{2}{2}$$

$$2/4$$

$$\underline{n=6} \rightarrow \sqrt{6} = \underline{\underline{2}}$$

$$6/2$$

$$1/6 \rightarrow 2 * 1 * \cancel{6} = 12$$

$$2/3 \rightarrow$$

$$\begin{array}{l} 1 \rightarrow n/1 \\ 2 \rightarrow n/2 \\ 3 \rightarrow n/3 \end{array}$$

$$\frac{2 \left[\frac{n}{2} \right] \textcircled{+}}{\left[\frac{n}{2} \right] \textcircled{2}}$$

$$\left(2 * \frac{n}{2} \right) * 2$$

$$4 \left(2 * 2 * \frac{n}{2} \right)$$

$$\textcircled{n=6}^2$$

$$\sqrt{n=2}^2$$

$$\frac{3}{4}$$

$$\frac{n}{3}$$

$$2 * \left\lceil \frac{n}{1} \right\rceil * 1 \leftarrow 1 \rightarrow n / n \rightarrow 1 \rightarrow \textcircled{2}^2$$

$$2 * \left\lceil \frac{n}{2} \right\rceil * 2 \quad 2 \rightarrow n/2 \quad n/2 \rightarrow 2 \rightarrow \textcircled{2}^2$$

$$2 * \left\lceil \frac{n}{3} \right\rceil * 3 \quad 3 \rightarrow n/3$$

$$\textcircled{\sqrt{n}}$$

$$\left\lceil \frac{n}{4} \right\rceil = 1 \quad n/5 \quad n/6 \rightarrow 1$$

$$1 \rightarrow n / n \rightarrow 1$$

$$2 \rightarrow n/2 = 3 / 3 \rightarrow n/3 = 2$$

$$\frac{1}{6} \quad \frac{2}{3} \quad \boxed{\frac{4}{5}} \rightarrow 2$$

$$\textcircled{1}$$

$n = (4)$
|
leftover
↓ ↓
①
=

$$2 * \left[\frac{6}{1} \right] * 1 \rightarrow 1/6 \rightarrow 2$$

$$2 * \left[\frac{6}{2} \right] * 2 \rightarrow 2/3 \rightarrow 2$$



